In[•]:=
```
Remove["Global`*"]
```
去除

```
ClearAll["Global`*"]
```
清除全部

```
Γself[a_, b_, c_] := Γ^a_"b.c"
dΓ[a_, b_, c_, d_] := Γ^a_"b.c,d"
Rself[a_, b_, c_, d_] := R^"_"a.b.c"d"
```

```
zuobiao = {t, x, y, z}
```

```
g0 = {-1 / Sqrt[z], z, z, 1 / Sqrt[z]}
g = DiagonalMatrix[g0]
```
对角矩阵

```
dimg = Dimensions[g][[1]]
```
维数

```
ing = Inverse[g]
```
逆

```
cΓself[a_, b_, c_] := Block[{μ = Position[zuobiao, a][[1, 1]],
```
块 位置

```
    ν = Position[zuobiao, b][[1, 1]], σ = Position[zuobiao, c][[1, 1]], k},
```
位置 位置

```
  Simplify[Sum[(1 / 2) * ing[[μ, k]] *
```
化简 求和

```
      (D[g[[ν, k]], c] + D[g[[σ, k]], b] - D[g[[σ, ν]], zuobiao[[k]]]), {k, dimg}]]]
```
偏导 偏导 偏导

```
Table[Γself[μ, ν, σ] == cΓself[μ, ν, σ],
```
表格

```
  {μ, zuobiao}, {ν, zuobiao}, {σ, zuobiao}] // TableForm
```
表格形式

```
cdΓ[a_, b_, c_, d_] := D[cΓself[a, b, c], d]
```
偏导

```
cR[a_, b_, c_, d_] :=
 cdΓ[d, a, c, b] - cdΓ[d, b, c, a] + Sum[cΓself[λ, c, a] * cΓself[d, b, λ], {λ, zuobiao}] -
```
求和

```
  Sum[cΓself[λ, c, b] * cΓself[d, a, λ], {λ, zuobiao}]
```
求和

```
Table[Rself[μ, ν, σ, ρ] == Simplify[cR[μ, ν, σ, ρ]],
```
表格 化简

```
  {μ, zuobiao}, {ν, zuobiao}, {σ, zuobiao}, {ρ, zuobiao}] // TableForm
```
表格形式

Out[•]=
```
{t, x, y, z}
```

Out[•]=
$$\left\{-\frac{1}{\sqrt{z}}, z, z, \frac{1}{\sqrt{z}}\right\}$$

Out[•]=

$$\left\{\left\{-\frac{1}{\sqrt{z}}, 0, 0, 0\right\}, \{0, z, 0, 0\}, \{0, 0, z, 0\}, \left\{0, 0, 0, \frac{1}{\sqrt{z}}\right\}\right\}$$

Out[•]=

4

Out[•]=

$$\left\{\left\{-\sqrt{z}, 0, 0, 0\right\}, \left\{0, \frac{1}{z}, 0, 0\right\}, \left\{0, 0, \frac{1}{z}, 0\right\}, \left\{0, 0, 0, \sqrt{z}\right\}\right\}$$

Out[•]//TableForm=

$\Gamma^{t}_{t.t} == 0$	$\Gamma^{t}_{x.t} == 0$	$\Gamma^{t}_{y.t} == 0$	$\Gamma^{t}_{z.t} == -\frac{1}{4z}$
$\Gamma^{t}_{t.x} == 0$	$\Gamma^{t}_{x.x} == 0$	$\Gamma^{t}_{y.x} == 0$	$\Gamma^{t}_{z.x} == 0$
$\Gamma^{t}_{t.y} == 0$	$\Gamma^{t}_{x.y} == 0$	$\Gamma^{t}_{y.y} == 0$	$\Gamma^{t}_{z.y} == 0$
$\Gamma^{t}_{t.z} == -\frac{1}{4z}$	$\Gamma^{t}_{x.z} == 0$	$\Gamma^{t}_{y.z} == 0$	$\Gamma^{t}_{z.z} == 0$
$\Gamma^{x}_{t.t} == 0$	$\Gamma^{x}_{x.t} == 0$	$\Gamma^{x}_{y.t} == 0$	$\Gamma^{x}_{z.t} == 0$
$\Gamma^{x}_{t.x} == 0$	$\Gamma^{x}_{x.x} == 0$	$\Gamma^{x}_{y.x} == 0$	$\Gamma^{x}_{z.x} == \frac{1}{2z}$
$\Gamma^{x}_{t.y} == 0$	$\Gamma^{x}_{x.y} == 0$	$\Gamma^{x}_{y.y} == 0$	$\Gamma^{x}_{z.y} == 0$
$\Gamma^{x}_{t.z} == 0$	$\Gamma^{x}_{x.z} == \frac{1}{2z}$	$\Gamma^{x}_{y.z} == 0$	$\Gamma^{x}_{z.z} == 0$
$\Gamma^{y}_{t.t} == 0$	$\Gamma^{y}_{x.t} == 0$	$\Gamma^{y}_{y.t} == 0$	$\Gamma^{y}_{z.t} == 0$
$\Gamma^{y}_{t.x} == 0$	$\Gamma^{y}_{x.x} == 0$	$\Gamma^{y}_{y.x} == 0$	$\Gamma^{y}_{z.x} == 0$
$\Gamma^{y}_{t.y} == 0$	$\Gamma^{y}_{x.y} == 0$	$\Gamma^{y}_{y.y} == 0$	$\Gamma^{y}_{z.y} == \frac{1}{2z}$
$\Gamma^{y}_{t.z} == 0$	$\Gamma^{y}_{x.z} == 0$	$\Gamma^{y}_{y.z} == \frac{1}{2z}$	$\Gamma^{y}_{z.z} == 0$
$\Gamma^{z}_{t.t} == -\frac{1}{4z}$	$\Gamma^{z}_{x.t} == 0$	$\Gamma^{z}_{y.t} == 0$	$\Gamma^{z}_{z.t} == 0$
$\Gamma^{z}_{t.x} == 0$	$\Gamma^{z}_{x.x} == -\frac{\sqrt{z}}{2}$	$\Gamma^{z}_{y.x} == 0$	$\Gamma^{z}_{z.x} == 0$
$\Gamma^{z}_{t.y} == 0$	$\Gamma^{z}_{x.y} == 0$	$\Gamma^{z}_{y.y} == -\frac{\sqrt{z}}{2}$	$\Gamma^{z}_{z.y} == 0$
$\Gamma^{z}_{t.z} == 0$	$\Gamma^{z}_{x.z} == 0$	$\Gamma^{z}_{y.z} == 0$	$\Gamma^{z}_{z.z} == -\frac{1}{4z}$

Out[•]//TableForm=

$R_{t.t.t}^{t} == 0$	$R_{t.t.t}^{x} == 0$	$R_{t.t.t}^{y} == 0$	$R_{t.t.t}^{z} == 0$
$R_{t.t.x}^{t} == 0$	$R_{t.t.x}^{x} == 0$	$R_{t.t.x}^{y} == 0$	$R_{t.t.x}^{z} == 0$
$R_{t.t.y}^{t} == 0$	$R_{t.t.y}^{x} == 0$	$R_{t.t.y}^{y} == 0$	$R_{t.t.y}^{z} == 0$
$R_{t.t.z}^{t} == 0$	$R_{t.t.z}^{x} == 0$	$R_{t.t.z}^{y} == 0$	$R_{t.t.z}^{z} == 0$
$R_{t.x.t}^{t} == 0$	$R_{t.x.t}^{x} == -\frac{1}{8z^2}$	$R_{t.x.t}^{y} == 0$	
$R_{t.x.x}^{t} == -\frac{1}{8\sqrt{z}}$	$R_{t.x.x}^{x} == 0$	$R_{t.x.x}^{y} == 0$	
$R_{t.x.y}^{t} == 0$	$R_{t.x.y}^{x} == 0$	$R_{t.x.y}^{y} == 0$	
$R_{t.x.z}^{t} == 0$	$R_{t.x.z}^{x} == 0$	$R_{t.x.z}^{y} == 0$	
$R_{x.t.t}^{t} == 0$	$R_{x.t.t}^{x} == \frac{1}{8z^2}$	$R_{x.t.t}^{y} == 0$	$R_{x.t.t}^{z} == 0$
$R_{x.t.x}^{t} == \frac{1}{8\sqrt{z}}$	$R_{x.t.x}^{x} == 0$	$R_{x.t.x}^{y} == 0$	$R_{x.t.x}^{z} == 0$
$R_{x.t.y}^{t} == 0$	$R_{x.t.y}^{x} == 0$	$R_{x.t.y}^{y} == 0$	$R_{x.t.y}^{z} == 0$
$R_{x.t.z}^{t} == 0$	$R_{x.t.z}^{x} == 0$	$R_{x.t.z}^{y} == 0$	$R_{x.t.z}^{z} == 0$
$R_{x.x.t}^{t} == 0$	$R_{x.x.t}^{x} == 0$	$R_{x.x.t}^{y} == 0$	$R_{x.x.t}^{z} =$
$R_{x.x.x}^{t} == 0$	$R_{x.x.x}^{x} == 0$	$R_{x.x.x}^{y} == 0$	$R_{x.x.x}^{z} =$
$R_{x.x.y}^{t} == 0$	$R_{x.x.y}^{x} == 0$	$R_{x.x.y}^{y} == 0$	$R_{x.x.y}^{z} =$
$R_{x.x.z}^{t} == 0$	$R_{x.x.z}^{x} == 0$	$R_{x.x.z}^{y} == 0$	$R_{x.x.z}^{z} =$
$R_{y.t.t}^{t} == 0$	$R_{y.t.t}^{x} == 0$	$R_{y.t.t}^{y} == \frac{1}{8z^2}$	$R_{y.t.t}^{z} == 0$
$R_{y.t.x}^{t} == 0$	$R_{y.t.x}^{x} == 0$	$R_{y.t.x}^{y} == 0$	$R_{y.t.x}^{z} == 0$
$R_{y.t.y}^{t} == \frac{1}{8\sqrt{z}}$	$R_{y.t.y}^{x} == 0$	$R_{y.t.y}^{y} == 0$	$R_{y.t.y}^{z} == 0$
$R_{y.t.z}^{t} == 0$	$R_{y.t.z}^{x} == 0$	$R_{y.t.z}^{y} == 0$	$R_{y.t.z}^{z} == 0$
$R_{y.x.t}^{t} == 0$	$R_{y.x.t}^{x} == 0$	$R_{y.x.t}^{y} == 0$	
$R_{y.x.x}^{t} == 0$	$R_{y.x.x}^{x} == 0$	$R_{y.x.x}^{y} == \frac{1}{4\sqrt{z}}$	
$R_{y.x.y}^{t} == 0$	$R_{y.x.y}^{x} == -\frac{1}{4\sqrt{z}}$	$R_{y.x.y}^{y} == 0$	
$R_{y.x.z}^{t} == 0$	$R_{y.x.z}^{x} == 0$	$R_{y.x.z}^{y} == 0$	
$R_{z.t.t}^{t} == 0$	$R_{z.t.t}^{x} == 0$	$R_{z.t.t}^{y} == 0$	$R_{z.t.t}^{z} == -\frac{1}{4z^2}$
$R_{z.t.x}^{t} == 0$	$R_{z.t.x}^{x} == 0$	$R_{z.t.x}^{y} == 0$	$R_{z.t.x}^{z} == 0$
$R_{z.t.y}^{t} == 0$	$R_{z.t.y}^{x} == 0$	$R_{z.t.y}^{y} == 0$	$R_{z.t.y}^{z} == 0$
$R_{z.t.z}^{t} == -\frac{1}{4z^2}$	$R_{z.t.z}^{x} == 0$	$R_{z.t.z}^{y} == 0$	$R_{z.t.z}^{z} == 0$
$R_{z.x.t}^{t} == 0$	$R_{z.x.t}^{x} == 0$	$R_{z.x.t}^{y} == 0$	$R_{z.x.t}$
$R_{z.x.x}^{t} == 0$	$R_{z.x.x}^{x} == 0$	$R_{z.x.x}^{y} == 0$	$R_{z.x.x}$
$R_{z.x.y}^{t} == 0$	$R_{z.x.y}^{x} == 0$	$R_{z.x.y}^{y} == 0$	$R_{z.x.y}$
$R_{z.x.z}^{t} == 0$	$R_{z.x.z}^{x} == \frac{1}{8z^2}$	$R_{z.x.z}^{y} == 0$	$R_{z.x.z}$